

Hardware Business Management System

Purpose of This Document

Defines offline behavior, local queue states, synchronization rules, and conflict handling.

Known Project Scope

- Mobile application and web administration system.
- Super Admin and Shop Admin roles.
- Products, purchases, sales, inventory and stock movement.
- Expenses, capital, debts, profit reporting, and 60/40 allocation support.
- Daily opening and daily closing workflows.
- Offline-first mobile storage using SQLite and central MySQL synchronization.
- Notifications, reports, audit trail, validation, error handling, security, and acceptance criteria.

Status Label Convention

Confirmed means the item is part of the agreed scope. Proposed means it is a recommended implementation detail that can still change. Pending Decision means the business owner must decide before a developer or Cloud AI treats it as final.

Status	Meaning	How to Treat It
Confirmed	Agreed project requirement or design direction.	Use as baseline for planning and implementation.
Proposed	Recommended approach that may still change.	Validate during design and owner review.
Pending Decision	Business, accounting, or technical rule not fully approved.	Do not hard code. Ask the owner or make configurable.

Offline First Design

The mobile app should save core shop records locally before depending on the server. This protects shop operations when internet access is unstable.

Sync State	Meaning	User Visibility	Status
Pending	Saved locally but not yet accepted by server.	Visible in sync status screen.	Confirmed
Syncing	Currently being sent to backend.	Visible as in progress.	Confirmed
Synced	Accepted by server and has server id.	Visible as complete.	Confirmed
Retry	Failed due to network or temporary issue.	User can retry or wait.	Confirmed
Rejected	Server validation failed.	User sees reason and correction path.	Confirmed
Conflict	Record conflicts with server state.	Requires configured or admin resolution.	Pending Decision

Sync Rules

- Every offline record must have local_uuid before first save.
- Backend must use local_uuid, user id, shop id, and device id where used to detect duplicate submissions.
- Sync must return accepted records, rejected records, retryable failures, and conflicts.
- Records must not be silently overwritten during conflicts.
- Failed sync must not delete local data.

Open Decisions That Must Not Be Invented

Decision Area	What Must Be Confirmed	Status
Profit formula	Exact formula for gross profit, net profit, cash profit, and expense treatment.	Pending Decision
Stock costing method	Latest cost, weighted average, FIFO, or another method for cost of goods sold.	Pending Decision
60/40 destination labels	Meaning of 60 percent and 40 percent amounts in business reports.	Pending Decision
Allocation timing	Daily, weekly, monthly, per closing, or owner-approved timing.	Pending Decision
Tax or VAT handling	Whether tax calculation is part of first release and how it affects reports.	Pending Decision
Returns refunds discounts	How returns, damaged goods, refunds, and discounts affect stock and profit.	Pending Decision

Acceptance of This Document

- The document is acceptable when the owner can identify which items are confirmed, proposed, and pending decision.
- A developer or Cloud AI can use the confirmed parts without needing to guess the project direction.
- Pending accounting rules remain visible and are not presented as final business rules.